

Om Shah



| +91 789 276 2884 | omamitshah@gmail.com | <http://linkedin.com/in/om-shah-2277b22b9/>
| <https://om-shah-portfolio.vercel.app/>

Profile

I am a Computer Engineering student at MPSTME, NMIMS University with a strong background in software development, machine learning, and full-stack web technologies. Skilled in building scalable systems and solving real-world problems using data-driven approaches. Passionate about AI and systems design, with a focus on continuous learning.

Education

MPSTME, NMIMS University B. Tech in Computer Engineering CGPA: 3.96/4	2022-2026
R.N. Podar School 12 th CBSE: 97%	2022
Sri Aurobindo Memorial School 10 th CBSE: 90.2%	2020

Projects

Reinforcement Learning for Portfolio Management through DDPG and PPO– August 2025

This project trains an autonomous agent to manage a stock portfolio using deep reinforcement learning (DRL). It implements two powerful algorithms, Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO), to learn an optimal trading strategy that aims to maximize financial returns.

Face detection using Deep Learning – May 2025

Built a deep learning system for face detection, implementing core concepts including CNN architecture design, custom data generators, loss functions, and training loops. Utilized WIDER FACE dataset.

PrediNator- Celebrity Predictor using Decision Tree– May 2025

A machine learning project that predicts celebrity names through a decision tree classifier that builds a decision tree based on QnA with the user. Implemented an algorithm to retrain the tree on new names of people that are not present in the database. Utilized Django Frontend and Backend

Neural Machine Translation of Spanish to English using Attention and Seq2Seq Architecture– July 2025

Designed and implemented a Spanish-to-English Neural Machine Translation (NMT) system using PyTorch with a Transformer-inspired sequence-to-sequence (Seq2Seq) model. Developed custom bidirectional encoder and attention-enabled decoder modules, incorporating multi-layer GRUs and context-aware attention mechanisms. Used BLEU score evaluation and mixed precision training for optimization.

Face detection using Deep Learning – May 2025

Built a deep learning system for face detection, implementing core concepts including CNN architecture design, custom data generators, loss functions, and training loops. Utilized WIDER FACE dataset.

Football video analysis system – March 2025

Developed a real-time tracking system that detects the player movements, speed, ball possession, and other statistics using Computer Vision, Machine Learning and integrated with a Django web interface.

Modern E-Commerce Website with React, Sanity, and Stripe- May 2025

Built a responsive e-commerce application using React with Sanity CMS for dynamic content management and integrated Stripe for secure real-time payment processing. Deployed on Vercel.

Dynamic Chat Room Application using MERN Stack – April 2024

Developed a full-stack chat application with user authentication using JWT Tokens. Utilized Mongo DB for storing user and message data, with Express and Node JS services for backend. Deployed as a dynamic site on Render.

Lung and brain tumour classification – November 2024

Created a Machine Learning based system to detect and classify tumours in lungs and brain if any using VGG16 model, Custom CNN models and Flask interface.

Certifications

- **Pursuing Honours from Coursera - 2022-2026**
 - **Deep Learning Specialization – Deeplearning.ai**
 - **Google Data Analytics – Google**
- **Quantum Machine Learning – EduXLabs, IIT-Madras – 2025**
- **IBM RAG and Agentic AI Professional Certificate – IBM – 2025**

Core Skills

Technical Skills – C++, Java, Python, HTML, CSS, JavaScript, MySQL, ReactJS, Node, Version Control, Flask, Computer Vision, Mongo DB, Neural Networks, machine Learning, Computer Networks, Data Structures and Algorithms, Chroma DB, RAG pipelines, multi-agentic systems.

Soft Skills – Leadership capability, team coordination, decision making, active listening, strategic planning, time management, goal setting.

Points of Responsibility and Volunteering

- **Technical Executive Role for the MPSTME's ACM committee for a year (2023-24)**
- **Class Representative for BTech CE division B**
- **Computer Tutor for underprivileged children (MS Word, Excel, Paint, Linux)**